

Gap Analysis



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1. Executive Summary

The ADHCRI Gap Analysis evaluates the current readiness of all ten ASEAN member states against the target capabilities required to implement a secure, interoperable, and climate-resilient regional digital health ecosystem.

The analysis focuses on digital infrastructure, internet connectivity, power resilience, healthcare funding capacity, workforce readiness, language accessibility, digital literacy, cybersecurity, and cross-border health data governance.

The findings show that ASEAN member states cannot be deployed using a single uniform approach. Instead, ADHCRI requires a phased readiness-based rollout using three deployment tiers: **Tier A – Advanced Readiness**, **Tier B – Mid-Infrastructure Readiness**, and **Tier C – Foundational Readiness**.

2. Gap Analysis Methodology

The gap analysis was conducted using the **ASEAN Digital Health Investment Readiness Framework (ADHIRF)**. Each member state was assessed against six readiness dimensions.

ADHIRF Dimension	Assessment Focus
Digital Infrastructure	Internet connectivity, digital health system maturity, interoperability readiness
Power Resilience	Electricity reliability, backup power, ability to support continuous healthcare operations
Healthcare Investment Capacity	Funding availability, sustainability, ability to support long-term platform operations
Digital Workforce Readiness	Digital literacy, training needs, ability to adopt new systems
Health Data Governance	Privacy, cybersecurity, data-sharing standards, sovereignty controls

Regional Collaboration Readiness	Ability to participate in coordinated cross-border public health intelligence
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Dataset Evidence Dashboard

To support the gap analysis and investment justification, a Power BI evidence dashboard was developed using the provided ASEAN health datasets. The dashboard compares health vulnerability, communicable disease burden, healthcare workforce capacity, and deployment readiness across ASEAN member states. This provides a data-backed view of why ADHCRI requires a phased implementation model rather than a one-size-fits-all regional rollout.

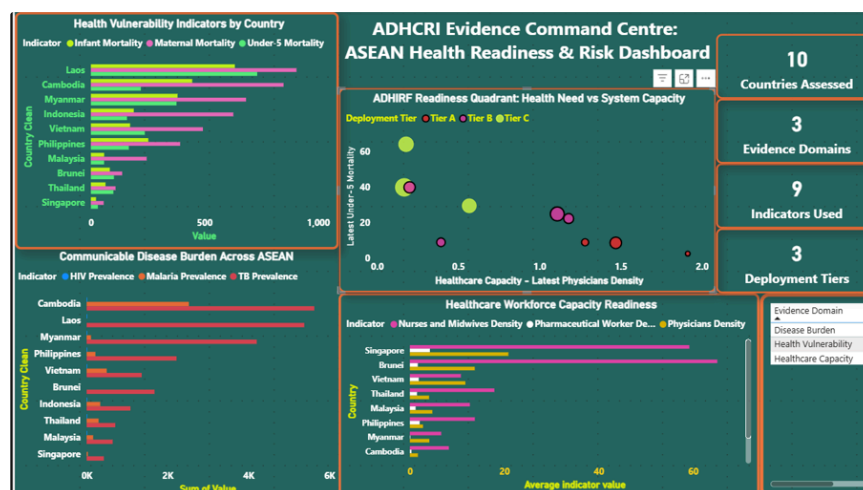


Figure: ADHCRI Evidence Command Centre showing dataset-backed evidence for health vulnerability, communicable disease burden, healthcare workforce capacity, and phased ASEAN deployment readiness.

3. Target-State Capability Model

Target Capability	Future-State Requirement
Secure Cross-Border Health Intelligence	Approved health intelligence can be shared securely across ASEAN member states.
Predictive Disease Surveillance	Public health agencies can detect abnormal disease patterns earlier.
Climate-Health Intelligence	Climate and geospatial indicators are linked to health risk planning.
Resilient Rural Healthcare Operations	Rural facilities can continue reporting during power or internet outages.
Trusted Data Governance	Countries retain control over sensitive health data while supporting approved regional intelligence sharing.
Inclusive Regional Adoption	Healthcare workers receive multilingual training and digital literacy support.
Phased Deployment	Countries are deployed based on readiness, not a one-size-fits-all

	rollout.
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4. Comprehensive Gap Analysis Matrix

Current State (As-Is)	Desired Future State (To-Be)	Identified Gap / Operational Friction	Proposed Action
Health data remains fragmented across hospitals, laboratories, clinics, and public health agencies.	Secure and standardized regional health intelligence exchange.	Limited interoperability delays disease intelligence and coordinated regional response.	Implement an API-first interoperability layer for approved regional health intelligence exchange.
Disease surveillance is largely reactive and dependent on inconsistent reporting.	Predictive and near-real-time disease surveillance.	Delayed outbreak detection increases containment costs and public health exposure.	Deploy predictive surveillance capabilities supported by AI and machine learning.
Climate and health information are managed separately by different agencies.	Integrated climate-health intelligence for public health planning.	Public health agencies lack visibility into climate-driven disease risk patterns.	Integrate GIS, climate indicators, and health surveillance data into decision dashboards.
Rural healthcare facilities experience unreliable internet and power supply.	Healthcare operations continue during infrastructure disruptions.	Connectivity and electricity failures interrupt reporting and service continuity.	Deploy edge nodes, offline synchronization, and backup power support in high-risk locations.
Healthcare funding and digital maturity vary significantly across ASEAN member states.	Readiness-based and equitable implementation.	A uniform rollout would increase implementation failure and widen digital inequality.	Apply ADHIRF tiering and phased deployment across Tier A, Tier B, and Tier C countries.
Cybersecurity and privacy standards differ across member states.	Trusted, governed, and auditable cross-border data exchange.	Inconsistent security controls reduce trust and increase cyber risk.	Establish regional data governance rules, role-based access, encryption, audit logs, and cybersecurity baseline controls.
Digital literacy and language needs vary across the regional workforce.	Inclusive adoption supported by localized training and multilingual guidance.	Language barriers and low digital literacy may reduce platform adoption.	Provide multilingual user support, country-level training, change champions, and adoption monitoring.
Cross-border health coordination relies heavily on manual communication and separate national processes.	Coordinated regional public health response supported by shared intelligence.	Slow information exchange limits timely response to outbreaks and climate-related health risks.	Establish shared dashboards, escalation workflows, and regional public health coordination protocols.

5. ASEAN Country Readiness Matrix

Regional Readiness & Gap Analysis - All 10 ASEAN Member States									
Scored 1 (weakest) to 5 (strongest) against ADHCRI target-state platform readiness. Sourced figures cited; scored fields are Lead BA judgement calibrated to sourced data.									
Country	Tier	Internet Penetration	Power Grid Reliability (1-5)	Health Exp. % GDP	Digital Health Maturity (1-5)	Primary Language(s)	Digital Literacy (1-5)	Overall Gap Score (1-5, 5=ready)	Transition Strategy
Singapore	A	92%*		5 3.7%*		4 English, Mandarin, Malay, Tamil	5		4.7 Anchor node for the region — integrate ADHCRI as an interoperability layer atop existing national health IT rather than a rebuild.
Brunei	A	97%*		5 2.9%*		3 Malay, English	4		4 Fast-track onboarding given small population and high per-capita spend (\$1,449*); use as an early proof-of-concept site.
Malaysia	A	97%*		4 4.0%*		4 Malay, English, Mandarin, Tamil	4		4 Leverage strong existing mobile infrastructure; focus transition effort on East Malaysia (Sabah/Sarawak) rural gap, not the peninsula.
Thailand	B	88%*		4 3.8%*		3 Thai	3		3.3 Build on existing Universal Coverage Scheme data infrastructure; Thai-language-first UI and training content.
Vietnam	B	86%*		3 4.6%*		3 Vietnamese	3		3 Strong mobile penetration (86%) offsets weaker fixed infrastructure; prioritize mobile-first clinic interfaces.
Indonesia	B	79%*		3 3.4%*		2 Bahasa Indonesia + regional languages	2		2.3 Archipelago geography is the core constraint — satellite backhaul (e.g., Starlink-class connectivity) required for outer-island facilities before hub go-live.
Philippines	B	18% fixed*/high mobile		3 5.1%*		3 Filipino, English + regional languages	3		3 Highest health-expenditure ratio in the region (5.1%) but fixed internet is the weak link — mobile-first, offline-sync edge nodes are mandatory, not optional.
Cambodia	C	81% (mobile)*		2 2.6% (est.)		2 Khmer	2		2 Community health worker network as the human bridge for digital literacy; Khmer-first training materials; edge nodes with extended offline buffering.
Laos	C	58% (mobile)*		2 1.33%*		1 Lao	1		1.3 Lowest health spend and internet penetration in the region — start with 2-3 pilot provinces, heaviest resilience-layer investment, longest phased timeline (Yr3-5).
Myanmar	C	45% (est.)		1 2.3% (est.)		1 Burmese + ethnic minority languages	1		1 Highest-risk deployment context — political and infrastructure instability; recommend phased, NGO-partnered pilot approach before full national hub commitment.
*Sourced: World Bank/WHO GHED, DataReport/We Are Social Oct 2025, ASEAN Investment healthcare brief, ERIA digital economy report (see Assumptions tab and citations in accompanying BRD). Unmarked figures are Lead BA estimates									

6. Tier-Based Transition Strategy

Deployment Tier	Countries	Transition Strategy
Tier A – Advanced Readiness	Singapore, Brunei, Malaysia	Begin early implementation, validate the platform, test governance rules, and establish centres of excellence.
Tier B – Mid-Infrastructure Readiness	Thailand, Vietnam, Indonesia, Philippines	Deploy after Tier A lessons are captured. Focus on interoperability, training, resilience, and regional integration.
Tier C – Foundational Readiness	Cambodia, Laos, Myanmar	Provide foundational support before full deployment, including connectivity, power resilience, workforce training, governance setup, and funding support.

7. Language and Digital Literacy Strategy

Barrier	Impact on ADHCRI	Transition Strategy
Multiple official and local languages across ASEAN	Users may struggle to adopt the platform if training and guidance are not localized.	Provide multilingual interfaces, translated training materials, and local-language support guides.
Uneven digital literacy among healthcare workers	Low adoption could reduce benefits realization and weaken data quality.	Deliver role-based training, hands-on onboarding, and digital literacy support before go-live.

Different levels of technology familiarity across countries	Some users may require more support than others during implementation.	Use country-level change champions and phased onboarding.
Limited rural technical support	Rural facilities may face delays resolving system or connectivity issues.	Provide local support teams, helpdesk escalation, offline guides, and simplified workflows.
Resistance to new ways of working	Manual processes may continue if users do not trust or understand the system.	Use stakeholder engagement, leadership sponsorship, and adoption tracking dashboards.

8. Key Findings and Recommendations

Finding	Meaning
ASEAN readiness is uneven.	Countries require different levels of support before deployment.
Infrastructure gaps create implementation risk.	Connectivity, power, and rural access must be addressed early.
Governance is central to trust.	Cross-border data exchange requires clear rules and sovereignty protections.
Digital literacy affects benefits realization.	Training and adoption support are essential to achieving ROI.
A phased rollout reduces risk.	Tier-based deployment is more realistic than a single regional launch.

Recommendations

Recommendation	Action
Use ADHIRF as the deployment readiness framework.	Assess countries before each implementation phase.
Start with Tier A countries.	Validate platform, governance, and operating model before wider rollout.
Expand to Tier B using lessons learned.	Strengthen interoperability, training, and infrastructure support.
Prepare Tier C before full deployment.	Invest in foundational infrastructure, power resilience, governance, and digital literacy.
Maintain regional governance oversight.	Use shared standards, risk reviews, and benefits tracking throughout implementation.

Closing Statement

The gap analysis confirms that ADHCRI is not only a technology implementation. It is a regional transformation programme that requires infrastructure readiness, governance alignment, workforce adoption, and equitable investment. A phased ADHIRF-based rollout provides the most practical path to sustainable implementation across all ten ASEAN member states.